Queensland Brain Institute

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Participant Information (Plain Language Statement)

Project Title: Measuring Episodic Memory with Virtual Reality

Investigators: Prof. Jason B. Mattingley, Mr Richard Ronayne, Mr Jayce Rushton, Professor Paul Dux, Professor Gail Robinson

1. Your Consent

You are invited to participate in this research project. This Plain Language Statement contains detailed information about the research project and the procedures involved. You will receive a copy of this statement. Please read this information carefully, and feel free to ask questions about the document or the project. Once you have read and have understood this document, you will be asked if you would like to participate. If you do, you will be required to sign a Statement of Informed Consent. Signing indicates you understand the provided information and that you give your consent to participate.

2. Purpose of the Project

Memory is a crucial component of our mental functioning. From remembering where you parked your car, to recognising a loved one's face, memory is the foundation from which our life's story can be built upon. As our ability to remember changes over the lifespan, accurate and valid measurement of memory performance is essential for both researchers and health professionals.

Memory is comprised of several domains, roughly defined by the type of information being recalled. Importantly, memory for one domain (such as people's faces) does not necessarily correlate with memory for other domains (such as remembering where you parked your car). Thus, it is important that each domain of memory be measured directly.

This research project aims to measure "episodic" memory. Episodic memory is the ability to recall and explicitly state details from past events. Traditionally this has been a difficult domain of memory to measure. Our past experiences are diverse, making direct comparisons between what two individuals can recall relatively meaningless and impossible to verify. If, however, both individuals had experienced identical events, and these events were known by the research team, it would be possible to directly compare recollection performance.

Virtual reality technology is one method of creating this scenario. By giving a large cohort of individuals the same lived experience, and then asking about specific details within that experience several days later, an individual's episodic memory performance can be directly compared against the average of the cohort. This is the core principle behind this project's experimental protocol.

The data collected in this research project will be used to make comparisons in a future study aimed at improving memory performance via non-invasive brain stimulation.

3. What you will be asked to do

By consenting to participate in this research project, you will be asked to provide some personal/health information to the research team, complete several cognitive testing tasks, view four virtual reality environments, and complete four memory tests based on your virtual reality experience. Participation is conducted both in-person and remotely. In-person participation will take approximately three and a half hours to complete, and the three remote participation sessions will take a further thirty minutes each to complete. This brings your full participation time to approximately five hours.

The personal information you will be required to provide includes your age, sex, education level, and previous virtual reality experience. The health information required includes any adverse experiences during virtual reality, if you have had a seizure, stroke, clinical EEG, head injury, illness resulting in brain injury, or other brain-related condition. You will also be asked if you suffer from headaches or take any medications.

The cognitive testing tasks include the National Adult Reading Test (NART), the Rey Auditory Verbal Learning Task (RAVLT), the Brief Visuospatial Memory Test-Revised (BVMT-R), and Raven's Progressive Matrices (RPM). These four tests are standard cognitive tests designed to measure several types of memory and cognition.

The virtual reality task requires you to use the research team's virtual reality headset to view four rooms in a virtual house. While viewing each room, you will be asked to describe several elements within the scene and pick up various "virtual" objects and move them from one location to another. Upon completion of all four virtual rooms, you will undertake one of four memory tests, each asking you to describe details of one of the virtual rooms. Upon finishing the first of four virtual reality memory tests, in-person participation is complete. The remaining three REMEMVR tests are to be completed remotely using the REMEMVR website.

4. Dissemination of Findings and Confidentiality

Data obtained from this project will be used to advance scientific knowledge concerning episodic memory and its relationship with other memory domains. We will not disclose any identifying information about you, or provided by you during the research, without your written permission or as required by law. Findings will be disseminated to the scientific community in the form of journal articles and conference presentations. No information that could identify you as a participant will be included. Fully anonymised data may be made available on public databases for use by other researchers, as required by granting bodies and/or publishers. At the end of the study a summary of the group findings will be made available to those participants who wish to know the outcome of the research. All data from this study will be stored securely in accordance with regulations prescribed by The University of Queensland. Data will be coded so that participants' identity is known only to the named investigators.

5. Benefit, Commitment, and Withdrawal

It should be recognised that it is not expected that you will receive direct health or other benefits from participation in this study. The data you provide, however, will help us to advance scientific understating of episodic memory and the factors that may affect memory storage and accuracy. You will receive $20 for each hour of participation. Payment will be made upon completion of your fourth REMEMVR memory test via direct bank transfer.

As your participation in this research is voluntary, you are free to withdraw from the study at any time, and for any reason. If you choose to withdraw you do not have to offer an explanation for your decision, and you may request that any unprocessed data collected from you be removed from the database. It should be noted that non-participation in the study, withdrawing from the study, or withholding of data, will in no way affect your relationship with The University of Queensland. For students of the University, withdrawal will not affect your grades.

6. Ethical Conduct of Research and Complaints

This study adheres to the Guidelines of the ethical review process of The University of Queensland and the National Statement on Ethical Conduct in Human Research. Whilst you are free to discuss your participation in this study with the researcher contactable at r.ronayne@uq.edu.au, if you would like to speak to an officer of the University not involved in the study, you may contact the Ethics Coordinator on 617 3365 3924 / 617 3443 1656 or email humanethics@research.uq.edu.au

Thank you for your time and interest in volunteering. Your effort is greatly appreciated.